

$$\textcircled{2} y = (x^2 - 1)^{3/2} (x^2 - 4)^{1/2}$$

$$\ln y = \frac{3}{2} \ln(x^2 - 1) + \frac{1}{2} \ln(x^2 - 4)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{3}{2} \times \frac{1}{x^2 - 1} \times 2x + \frac{1}{2} \times \frac{1}{x^2 - 4} \times 2x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{3x}{x^2 - 1} + \frac{x}{x^2 - 4}$$

$$\therefore \frac{dy}{dx} = y \times \left(\frac{3x}{x^2 - 1} + \frac{x}{x^2 - 4} \right)$$

$$= (x^2 - 1)^{3/2} (x^2 - 4)^{1/2} \left(\frac{3x}{x^2 - 1} + \frac{x}{x^2 - 4} \right) \text{ (Ans.)}$$

$$\textcircled{2} y = \frac{x^3 \sqrt{x^2 - 4}}{\sqrt{x^2 + 3}}$$

$$\ln y = 3 \ln x + \frac{1}{2} \ln(x^2 - 4) - \frac{1}{2} \ln(x^2 + 3)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{3}{x} + \frac{1}{2} \times \frac{1}{x^2 - 4} \times 2x - \frac{1}{2} \times \frac{1}{x^2 + 3} \times 2x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{3}{x} + \frac{x}{x^2 - 4} - \frac{x}{x^2 + 3}$$

$$\therefore \frac{dy}{dx} = y \times \left(\frac{3}{x} + \frac{x}{x^2 - 4} - \frac{x}{x^2 + 3} \right)$$

$$= \frac{x^3 \sqrt{x^2 - 4}}{\sqrt{x^2 + 3}} \times \left(\frac{3}{x} + \frac{x}{x^2 - 4} - \frac{x}{x^2 + 3} \right) \text{ (Ans.)}$$

$$61 \quad y = \frac{1-x}{(1-x^3)^{1/2}}$$

$$\ln y = \ln(1-x) - \frac{1}{2} \ln(1-x^3)$$

$$\frac{x^3 \ln x}{x-1} = 6 \quad (8)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{1-x} \times (-1) - \frac{1}{2} \times \frac{1}{1-x^3} \times (-3x^2) = 6 \quad (9)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{1-x} - \frac{3x^2}{2(1-x^3)} + \frac{1}{x} = \frac{6b}{xb} \cdot \frac{1}{6}$$

$$\therefore \frac{dy}{dx} = y \cdot x \left\{ \frac{3x^2}{2(1-x^3)} - \frac{1}{1-x} \right\} + \frac{1}{x} = \frac{6b}{xb} \cdot \frac{1}{6}$$

$$\left(\frac{1-x}{(1-x^3)^{1/2}} \times \left\{ \frac{3x^2}{2(1-x^3)} - \frac{1}{1-x} \right\} + \frac{1}{x} \right) = \frac{6b}{xb} \quad (\text{Ans.})$$

$$8) y = \frac{x \sin^{-1} x}{\sqrt{1-x^2}}$$

$$\ln y = \ln x + \ln \sin^{-1} x - \frac{1}{2} \ln(1-x^2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} + \frac{1}{\sin^{-1} x} \times \frac{1}{\sqrt{1-x^2}} - \frac{1}{2} \times \frac{1}{1-x^2} \times (-2x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} + \frac{1}{\sin^{-1} x \sqrt{1-x^2}} + \frac{x}{1-x^2}$$

$$\frac{dy}{dx} = y \times \left(\frac{1}{x} + \frac{1}{\sin^{-1} x \sqrt{1-x^2}} + \frac{x}{1-x^2} \right)$$

$$= \frac{x \sin^{-1} x}{\sqrt{1-x^2}} \times \left(\frac{1}{x} + \frac{1}{\sin^{-1} x \sqrt{1-x^2}} + \frac{x}{1-x^2} \right) \quad (A)$$

$$9) y = \frac{x \cos^{-1} x}{\sqrt{1-x^2}}$$

$$\ln y = \ln x + \ln \cos^{-1} x - \frac{1}{2} \ln(1-x^2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} + \frac{1}{\cos^{-1} x} \times \frac{1}{\sqrt{1-x^2}} - \frac{1}{2} \times \frac{1}{1-x^2} \times (-2x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} - \frac{1}{\cos^{-1} x \sqrt{1-x^2}} + \frac{x}{1-x^2}$$

$$\frac{dy}{dx} = y \times \left(\frac{1}{x} - \frac{1}{\cos^{-1} x \sqrt{1-x^2}} + \frac{x}{1-x^2} \right)$$

$$= \frac{x \cos^{-1} x}{\sqrt{1-x^2}} \times \left(\frac{1}{x} - \frac{1}{\cos^{-1} x \sqrt{1-x^2}} + \frac{x}{1-x^2} \right)$$

(Ans.)

$$\underline{Q} \quad y = \frac{x \ln x}{\sqrt{1+x^2}}$$

$$\ln y = \ln x + \ln \ln x + \frac{1}{2} \ln(1+x^2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} + \frac{1}{\ln x} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{1}{1+x^2} \cdot 2x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{1}{x} + \frac{1}{x \ln x} + \frac{x}{1+x^2}$$

$$\therefore \frac{dy}{dx} = y \cdot x \left(\frac{1}{x} + \frac{1}{x \ln x} + \frac{x}{1+x^2} \right)$$

$$= \frac{x \ln x}{\sqrt{1+x^2}} \cdot x \left(\frac{1}{x} + \frac{1}{x \ln x} + \frac{x}{1+x^2} \right) \quad (\text{Ans})$$

$$1) y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$$

$$\frac{y \cdot dy}{dx} = \frac{1}{2} \ln(1+x^2)$$

$$\ln y = \ln e^{x^2} + \ln \tan^{-1} x - \frac{1}{2} \ln(1+x^2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 2x + \frac{1}{\tan^{-1} x} \times \frac{1}{1+x^2} - \frac{1}{2} \times \frac{1}{1+x^2} (2x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 2x + \frac{1}{\tan^{-1} x (1+x^2)} - \frac{x}{1+x^2}$$

$$\therefore \frac{dy}{dx} = y \times \left(2x + \frac{1}{\tan^{-1} x (1+x^2)} - \frac{x}{1+x^2} \right)$$

$$= \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left(2x + \frac{1}{\tan^{-1} x (1+x^2)} - \frac{x}{1+x^2} \right) \text{Ans.}$$

$$2) y = (x^2+1)^{\tan^{-1} x} \ln(1+\sin^2 x)$$

$$\ln y = \tan^{-1} x \ln(x^2+1) + \ln \ln(1+\sin^2 x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \tan^{-1} x \times \frac{1}{1+x^2} \times 2x + \frac{1}{\ln(1+\sin^2 x)} \times 2 \sin x \times \cos x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{2x \tan^{-1} x}{1+x^2} + \frac{\sin 2x}{\ln(1+\sin^2 x)}$$

$$\therefore \frac{dy}{dx} = y \times \left(\frac{2x \tan^{-1} x}{1+x^2} + \frac{\sin 2x}{\ln(1+\sin^2 x)} \right)$$

$$= (x^2+1)^{\tan^{-1} x} \ln(1+\sin^2 x) \left(\frac{2x \tan^{-1} x}{1+x^2} + \frac{\sin 2x}{\ln(1+\sin^2 x)} \right) \text{Ans}$$

$$20] y = \left(\frac{x}{1 + \sqrt{1-x^2}} \right)^4$$

$$\ln y = 4 \ln x - 4 \ln 1 - \frac{4}{2} \ln(1-x^2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{4}{x} - 0 - 2 \cdot \frac{1}{1-x^2} \times (-2x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{4}{x} + \frac{4x}{1-x^2}$$

$$\therefore \frac{dy}{dx} = y \times \left(\frac{4-4x^2+4x^2}{x(1-x^2)} \right)$$

$$= \left(\frac{x}{1+\sqrt{1-x^2}} \right)^4 \times \frac{4}{x(1-x^2)}$$

Ans.

$$20] y = \left(\frac{x}{1 + \sqrt{1-x^2}} \right)^4$$

$$\ln y = 4 \ln x - 4 \ln(1 + \sqrt{1-x^2})$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{4}{x} - \frac{4}{1+\sqrt{1-x^2}} \times \left[0 + \frac{1}{2\sqrt{1-x^2}} \times (-2x) \right]$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{4}{x} + \frac{4}{1+\sqrt{1-x^2}} \times \frac{x}{\sqrt{1-x^2}}$$

$$\therefore \frac{dy}{dx} = y \cdot 4 \left[\frac{1}{x} + \frac{x}{\sqrt{1-x^2}(1+\sqrt{1-x^2})} \right] \text{ Ans.}$$

$$21) y = \left[\frac{x}{x + \sqrt{a^2 - x^2}} \right]^n$$

$$\ln y = n \ln x - n \ln (x + \sqrt{a^2 - x^2})$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{n}{x} - \frac{n}{x + \sqrt{a^2 - x^2}} \left\{ 1 + \frac{1}{2\sqrt{a^2 - x^2}} x (-2x) \right\}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{n}{x} - \frac{n}{x + \sqrt{a^2 - x^2}} \times \left(1 - \frac{x}{\sqrt{a^2 - x^2}} \right)$$

$$\frac{dy}{dx} = y \cdot n \left[\frac{1}{x} - \frac{1}{x + \sqrt{a^2 - x^2}} \times \left(1 - \frac{x}{\sqrt{a^2 - x^2}} \right) \right]$$

Ans.

$$22) y = x^{\sin x}$$

$$\ln y = \sin x \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \left(\sin x \cdot \frac{1}{x} \right) + \ln x \cdot \cos x$$

~~$$\frac{dy}{dx} = \frac{\sin x}{x} \times y = \frac{\sin x}{x} \times x^{\sin x} = \frac{\sin x}{x} x^{\sin x} \quad (\text{Ans})$$~~

$$\frac{dy}{dx} = y \times \left(\frac{\sin x}{x} + \cos x \ln x \right)$$

$$= x^{\sin x} \left(\frac{\sin x}{x} + \cos x \ln x \right) \quad (\text{Ans})$$

$$26] \quad y = x^{\cos^{-1} x}$$

$$\ln y = \cos^{-1} x \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \cos^{-1} x \times \frac{1}{x} + \ln x \cdot -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = y \times \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$$

$$= \frac{x^{\cos^{-1} x}}{x} \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right] \quad (\text{Ans})$$

$$28] \quad y = x^{\sin^{-1} x}$$

$$\ln y = \sin^{-1} x \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\sin^{-1} x}{x} + \ln x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = x^{\sin^{-1} x} \left[\frac{\sin^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}} \right] \quad (\text{Ans})$$

$$29] \quad y = (\sin x)^{\cos x}$$

$$\ln y = \cos x \ln \sin x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\cos x}{\cos x} + \frac{\ln \sin x}{\cos x} \times (-\sin x)$$

$$\frac{dy}{dx} = (\sin x)^{\cos x} (1 - \sin x \ln \sin x)$$

23) $y = (\sin x)^{\cos x}$

$\ln y = \cos x \ln \sin x$

$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\cos x}{\sin x} \cdot \cos x + \ln \sin x \cdot (-\sin x)$

$\frac{dy}{dx} = y \times [\cos x \cot x - \sin x \ln \sin x]$

$(Ans) = (\sin x)^{\cos x} [\cos x \cot x - \sin x \ln \sin x]$

24) $y = (\sin x)^{\tan x}$

$\ln y = \tan x \ln \sin x$

$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\tan x}{\sin x} \cdot \cos x + \ln \sin x \cdot \sec x$

$\frac{dy}{dx} = y \cdot [\tan x \cot x + \sec x \ln \sin x]$

$= (\sin x)^{\tan x} [1 + \sec x \ln \sin x]$

(Ans)

29] $y = (\sin x)^{\cos^{-1} x}$

$$\ln y = \cos^{-1} x \ln \sin x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\cos^{-1} x}{\sin x} \times \cos x + \ln \sin x \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = y \times \left[\cot x \cos^{-1} x - \frac{\ln \sin x}{\sqrt{1-x^2}} \right]$$

$$= (\sin x)^{\cos^{-1} x} \left[\cot x \cos^{-1} x - \frac{\ln \sin x}{\sqrt{1-x^2}} \right] \quad (\text{Ans})$$

30] $y = (\sin^{-1} x)^{\ln x}$

$$\ln y = \ln x \ln \sin^{-1} x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\ln x}{\sin^{-1} x} \times \frac{1}{\sqrt{1-x^2}} + \ln \sin^{-1} x \cdot \frac{1}{x}$$

$$\frac{dy}{dx} = (\sin^{-1} x)^{\ln x} \left[\frac{\ln x}{\sin^{-1} x \sqrt{1-x^2}} + \frac{\ln \sin^{-1} x}{x} \right] \quad (\text{Ans})$$

31] $y = (\tan^{-1} x)^{\sin x + \cos x}$

$$\ln y = (\sin x + \cos x) \ln \tan^{-1} x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\sin x + \cos x}{\tan^{-1} x} \cdot \frac{1}{1+x^2} + \ln \tan^{-1} x (\cos x - \sin x)$$

$$\frac{dy}{dx} = y \times \left[\frac{\sin x + \cos x}{\tan^{-1} x (1+x^2)} + \ln \tan^{-1} x (\cos x - \sin x) \right]$$

$$= (\tan^{-1} x)^{\sin x + \cos x} \left[\frac{\sin x + \cos x}{\tan^{-1} x (1+x^2)} + \ln \tan^{-1} x (\cos x - \sin x) \right] \quad (\text{Ans})$$

$$20) y = (x^2 + 2)^{2-x}$$

$$\ln y = (2-x) \ln(x^2 + 2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{2-x}{x^2+2} \times 2x + \ln(x^2+2) \times (-1)$$

$$\frac{dy}{dx} = y \left[\frac{2x(2-x)}{x^2+2} - \ln(x^2+2) \right]$$

$$= (x^2+2)^{2-x} \left[\frac{(2-x)2x}{x^2+2} - \ln(x^2+2) \right] \text{ Ans.}$$

$$21) y = \left(\frac{3+x}{5+x} \right)^{8+2x}$$

$$\ln y = (8+2x) \{ \ln(3+x) - \ln(5+x) \}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = (8+2x) \left\{ \frac{1}{3+x} \times 1 - \frac{1}{5+x} \times 1 \right\} + 2 \{ \ln(3+x) - \ln(5+x) \}$$

$$\frac{dy}{dx} = 2y \left(\frac{1}{3+x} - \frac{1}{5+x} \right)$$

$$= 2 \left(\frac{3+x}{5+x} \right)^{8+2x} \left(\frac{1}{3+x} - \frac{1}{5+x} \right) \text{ (Ans)}$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = (8+2x) \left(\frac{1}{3+x} - \frac{1}{5+x} \right) + 2 \{ \ln(3+x) - \ln(5+x) \}$$

$$\frac{dy}{dx} = \left(\frac{3+x}{5+x} \right)^{8+2x} \times 2 \left\{ \frac{4+x}{3+x} - \frac{4+x}{5+x} + \ln \frac{3+x}{5+x} \right\}$$

(Ans)

$$22] y = e^{\cot^{-1} x}$$

$$\ln y = \cot^{-1} x \cdot \ln e$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \cot^{-1} x \cdot 1 + \ln e - \frac{1}{1+x^2}$$

$$22] y = e^{\cot^{-1} x}$$

$$\ln y = \cot^{-1} x \cdot \ln e$$

$$\ln y = \cot^{-1} x \cdot 1$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = - \frac{1}{1+x^2}$$

$$\therefore \frac{dy}{dx} = - \frac{e^{\cot^{-1} x}}{1+x^2} \quad (\text{Ans})$$

$$23] y = e^{(\sin^{-1} x)^2}$$

$$\ln y = 2(\sin^{-1} x) \ln e = 2(\sin^{-1} x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1-x^2}} \times \sin^{-1} x$$

$$\frac{dy}{dx} = \frac{2 \cdot e^{(\sin^{-1} x)^2} \sin^{-1} x}{\sqrt{1-x^2}} \quad (\text{Ans})$$

$$(24)] \quad \left(\frac{x^2 + 1}{x} + x \log x \right) \quad (\text{Ans})$$

$$28] y = 3^{\sqrt{1+x+x^2}}$$

$$\ln y = \sqrt{1+x+x^2} \ln 3$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \ln 3 \cdot \frac{1}{2\sqrt{1+x+x^2}} \times (1+2x) + 0$$

$$\frac{dy}{dx} = 3^{\sqrt{1+x+x^2}} \times \frac{(1+2x) \ln 3}{2\sqrt{1+x+x^2}} \quad (\text{Ans})$$

$$29] y = (\sin x)^{\ln x} + \sin^2 \cos^{-1} x$$

$$\text{Let, } u = (\sin x)^{\ln x}$$

$$\ln u = \ln x \ln \sin x$$

$$\therefore \frac{1}{u} \cdot \frac{du}{dx} = \frac{\ln x}{\sin x} \cdot \cos x + \frac{\ln \sin x}{x}$$

$$\therefore \frac{du}{dx} = (\sin x)^{\ln x} \left[\ln x \cdot \cot x + \frac{\ln \sin x}{x} \right] \quad \text{--- (1)}$$

$$\text{Again, } v = \sin^2 \cos^{-1} x$$

$$\frac{dv}{dx} = 2 \sin \cos^{-1} x \cdot \cos \cos^{-1} x - \frac{1}{\sqrt{1-x^2}}$$

$$= -2x \frac{\sin \cos^{-1} x}{\sqrt{1-x^2}}$$

$$= -2x \frac{\sin \sin^{-1} \sqrt{1-x^2}}{\sqrt{1-x^2}} = -2x \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}} = -2x$$

$$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$= (\sin x)^{\ln x} \left[\ln x \cdot \cot x + \frac{\ln \sin x}{x} \right] - 2x$$

(Ans.)

$$20) y = 10^{\ln \sin x}$$

$$\ln y = \ln \sin x \ln 10$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{\ln 10}{\sin x} \cos x + 0$$

$$\frac{dy}{dx} = 10^{\ln \sin x} \cdot \cot x \ln 10 \quad (\text{Ans.})$$

$$21) y = x^{x^x}$$

$$\ln y = x^x \ln x$$

$$\ln(\ln y) = x \ln x + \ln \ln x$$

$$\frac{1}{\ln y} \times \frac{1}{y} \cdot \frac{dy}{dx} = \frac{x}{x} + \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$\frac{dy}{dx} = y \ln y \left(1 + \frac{1}{x \ln x} \right)$$

$$= x^{x^x} x^x \ln x \left(1 + \frac{1}{x \ln x} \right)$$

$$= x^{x^x} x^x \left[\frac{\ln x \left(\frac{x \ln x + 1}{x \ln x} \right)}{x} \right]$$

$$= x^{x^x} x^x \left(\frac{1 + x \ln x}{x} \right) \quad (\text{Ans.})$$

$$26] y = (\sin \sqrt{1+x^2})^{\ln \cos x}$$

$$\ln y = \ln \cos x \ln \sin \sqrt{1+x^2}$$

$$\frac{1}{y} \cdot \ln y = \frac{\ln \cos x}{\sin \sqrt{1+x^2}} \times \cos \sqrt{1+x^2} \times \frac{1}{2\sqrt{1+x^2}} \times 2x + \frac{\ln \sin \sqrt{1+x^2}}{\cos x} \times -\sin x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{x \ln \cos x \cot \sqrt{1+x^2}}{\sqrt{1+x^2}} - \tan x \ln \sin \sqrt{1+x^2}$$

$$\frac{dy}{dx} = (\sin \sqrt{1+x^2})^{\ln \cos x} \left[\frac{x \ln \cos x \cot \sqrt{1+x^2}}{\sqrt{1+x^2}} - \tan x \ln \sin \sqrt{1+x^2} \right]$$

Ans.

$$27] y = (2^x + x^2)^{f(x)}$$

$$\ln y = f(x) \ln (2^x + x^2)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{f(x)}{2^x + x^2} \times (2^x \ln 2 + 2x) + \ln (2^x + x^2) f'(x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = y \times \left[\frac{f(x) (2^x \ln 2 + 2x)}{2^x + x^2} + f'(x) \ln (2^x + x^2) \right]$$

$$= (2^x + x^2)^{f(x)} \times \left[\frac{f(x) (2^x \ln 2 + 2x)}{2^x + x^2} + f'(x) \ln (2^x + x^2) \right]$$

Ans

29) $y = x^{y^x}$

$\ln y = y^x \ln x$

$\ln \ln y = x \ln y + \ln \ln x$

$\frac{1}{y \ln y} \cdot \frac{dy}{dx} = \frac{x}{y} \frac{dy}{dx} + \ln y + \frac{1}{x \ln x}$

$\left(\frac{1}{y \ln y} - \frac{x}{y} \right) \frac{dy}{dx} = \ln y + \frac{1}{x \ln x}$

$\left(\frac{1 - x \ln y}{y \ln y} \right) \frac{dy}{dx} = \frac{\ln y x \ln x + 1}{x \ln x}$

$\therefore \frac{dy}{dx} = \frac{(\ln y x \ln x + 1) y \ln y}{x \ln x (1 - x \ln y)} \quad (\text{Ans.})$

30) $y = x^x + x^{1/x}$

$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

$= x^x (1 + \ln x) + x^{1/x} \cdot \frac{1}{x^2} (1 - \ln x)$
(Ans.)

Let, $u = x^x$
 $\ln u = x \ln x$

$\frac{1}{u} \cdot \frac{du}{dx} = \frac{x}{x} + \ln x$

$\frac{du}{dx} = x^x (1 + \ln x)$

Again, $v = x^{1/x}$

$\ln v = \frac{1}{x} \ln x$

$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{1}{x} \times \frac{1}{x} + \ln x \cdot -\frac{1}{x^2}$

$\frac{dv}{dx} = x^{1/x} \cdot \frac{1}{x^2} (1 - \ln x)$

soln

$$y = \left(1 + \frac{1}{x}\right)^x + x^{1/x}$$

Again, $v = x^{1/x}$

$$\ln v = \frac{1}{x} \ln x$$

$$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{1}{x^2} + \ln x \left(-\frac{1}{x^2}\right) \quad \frac{dv}{dx} = \left(1 + \frac{1}{x}\right)^x \left[\ln\left(1 + \frac{1}{x}\right) - \frac{1}{1+x}\right]$$

$$\frac{dv}{dx} = x^{1/x} \left(\frac{1}{x^2} - \frac{1}{x^2} \ln x\right) \quad \text{--- (ii)}$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \\ &= \left(1 + \frac{1}{x}\right)^x \left[\ln\left(1 + \frac{1}{x}\right) - \frac{1}{1+x}\right] + x^{1/x} \left(\frac{1}{x^2} - \frac{1}{x^2} \ln x\right) \end{aligned} \quad \text{(Ans.)}$$

soln

$$y = x^{\tan x} + (\sin x)^{\cos x}$$

let, $u = x^{\tan x}$

$$\ln u = \tan x \ln x$$

$$\frac{1}{u} \cdot \frac{du}{dx} = \frac{\tan x}{x} + \ln x \sec^2 x$$

$$\frac{du}{dx} = x^{\tan x} \left[\frac{\tan x}{x} + \ln x \sec^2 x \right]$$

Again, $v = (\sin x)^{\cos x}$

$$\ln v = \cos x \ln \sin x$$

$$\frac{1}{v} \frac{dv}{dx} = \frac{\cos x}{\sin x} \cdot \cos x + \ln \sin x (-\sin x)$$

$$\frac{dv}{dx} = (\sin x)^{\cos x} [\cos x \cot x - \sin x \ln \sin x]$$

$$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$= x^{\tan x} \left[\frac{\tan x}{x} + \ln x \sec^2 x \right] + (\sin x)^{\cos x} [\cos x \cot x - \sin x \ln \sin x]$$

(Ans)

Q2] $y = x^{\ln x} + x^{\cos^{-1} x}$

$$u = x^{\ln x}$$

$$\ln u = \ln x \ln x$$

$$\frac{1}{u} \frac{du}{dx} = \frac{\ln x}{x} + \frac{\ln x}{x}$$

$$\frac{du}{dx} = x^{\ln x} \frac{2 \ln x}{x}$$

— (1)

$$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$= x^{\ln x} \frac{2 \ln x}{x} + x^{\cos^{-1} x} \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$$

(Ans)

$$\ln v = \cos^{-1} x \ln x$$

$$\frac{1}{v} \frac{dv}{dx} = \frac{\cos^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}}$$

$$\frac{dv}{dx} = x^{\cos^{-1} x} \left[\frac{\cos^{-1} x}{x} - \frac{\ln x}{\sqrt{1-x^2}} \right]$$

Q2) $y = x^x + (\sin x)^{\ln x}$

$u = x^x$

$\ln u = x \ln x$

$\frac{1}{u} \cdot \frac{du}{dx} = \frac{x}{x} + \ln x$

$\frac{du}{dx} = x^x (1 + \ln x)$

$v = (\sin x)^{\ln x}$

$\ln v = \ln x \ln \sin x$

$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{\ln x}{\sin x} \cdot \cos x + \ln \sin x \cdot \frac{1}{x}$

$\frac{dv}{dx} = (\sin x)^{\ln x} \left[\cot x \ln x + \frac{\ln \sin x}{x} \right]$

$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

$= x^x (1 + \ln x) + (\sin x)^{\ln x} \left[\cot x \ln x + \frac{\ln \sin x}{x} \right] \text{ (Ans.)}$

Q3) $y = x \cos^{-1} x + (\sin x)^{\ln x}$

$u = x \cos^{-1} x$

$\ln u = \cos^{-1} x \ln x$

$\frac{1}{u} \cdot \frac{du}{dx} = \frac{\cos^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}}$

$\frac{du}{dx} = x \cos^{-1} x \left[\frac{\cos^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}} \right]$

$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

$= x \cos^{-1} x \left[\frac{\cos^{-1} x}{x} + \frac{\ln x}{\sqrt{1-x^2}} \right] + (\sin x)^{\ln x} \left[\cot x \ln x + \frac{\ln \sin x}{x} \right]$

(Ans.)

Q81 $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$

$u = \sin x^{\cos x}$

$\ln u = \cos x \ln \sin x$

$\frac{1}{u} \cdot \frac{du}{dx} = \frac{\cos x}{\sin x} \times \cos x + \ln \sin x (-\cos x)$

$\frac{du}{dx} = \sin x^{\cos x} [\cot x \cos x - \sin x \ln \sin x]$

$v = \cos x^{\sin x}$

$\ln v = \sin x \ln \cos x$

$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{\sin x}{\cos x} \cdot (-\sin x) + \ln \cos x \cdot \cos x$

$\frac{dv}{dx} = \cos x^{\sin x} [\cos x \ln \cos x - \tan x \sin x]$

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

$= \sin x^{\cos x} [\cos x \cot x - \sin x \ln \sin x] + \cos x^{\sin x} [\cos x \ln \cos x - \sin x \tan x]$
(Ans)

Q82 $y = \sin x^{\cos x} + \tan x^{\sin x}$

$v = \tan x^{\sin x}$

$\ln v = \sin x \ln \tan x$

$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{\sin x}{\tan x} \cdot \frac{1}{1+x^2} + \cos x \ln \tan x$

$\frac{dv}{dx} = \tan x^{\sin x} \left[\frac{\cos x}{1+x^2} + \cos x \ln \tan x \right]$

$\frac{dy}{dx} = \sin x^{\cos x} [\cot x \cos x - \sin x \ln \sin x] + \tan^{\sin x} \left[\frac{\cos x}{1+x^2} + \cos x \ln \tan x \right]$
(Ans)

$$\underline{Q4} \quad y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$$

$$u = \tan x^{\cot x}$$

$$\ln u = \cot x \ln \tan x$$

$$\frac{1}{u} \cdot \frac{du}{dx} = \frac{\cot x}{\tan x} \cdot \frac{1}{1+x^2} - \frac{\ln \tan x}{1+x^2}$$

$$\cancel{v = \cot x}$$

$$\frac{du}{dx} = \tan x^{\cot x} \left[\frac{\cot^2 x}{1+x^2} - \frac{\ln \tan x}{1+x^2} \right]$$

$$[v = \cot x]$$

$$\ln v = \tan x \ln \cot x$$

$$\frac{1}{v} \cdot \frac{dv}{dx} = \frac{\tan x}{\cot x} \left(-\frac{1}{1+x^2} \right) + \frac{\ln \cot x}{1+x^2}$$

$$\frac{dv}{dx} = \cot x \tan x \left[\frac{\ln \cot x}{1+x^2} - \frac{\tan^2 x}{1+x^2} \right]$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$= \tan x^{\cot x} \left[\frac{\cot^2 x}{1+x^2} - \frac{\ln \tan x}{1+x^2} \right] + \cot x^{\tan x} \left[\frac{\ln \cot x}{1+x^2} - \frac{\tan^2 x}{1+x^2} \right]$$

$$9) y = \frac{x \cos 2x}{e^{3x} \sqrt{1+x^2}}$$

$$\ln y = \ln x + \ln \cos 2x - \ln e^{3x} - \frac{1}{2} \ln(1+x^2)$$

$$\frac{1}{y} \times \frac{dy}{dx} = \frac{1}{x} + \frac{1}{\cos 2x} (-2 \sin 2x) - 3 - \frac{1}{2} \times \frac{2x}{1+x^2}$$

$$\frac{1}{y} \times \frac{dy}{dx} = \frac{1}{x} - 2 \tan 2x - 3 - \frac{x}{1+x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x \cos 2x}{e^{3x} \sqrt{1+x^2}} \left[\frac{1}{x} - 2 \tan 2x - 3 - \frac{x}{1+x^2} \right]$$

$$10) y = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}}$$

$$\ln y = x^2 \ln e + \ln \tan^{-1} x - \frac{1}{2} \ln(1+x^2)$$

$$\frac{1}{y} \times \frac{dy}{dx} = 2x + \frac{1}{\tan^{-1} x} \times \frac{1}{1+x^2} - \frac{1}{2} \times \frac{2x}{1+x^2}$$

$$\frac{1}{y} \times \frac{dy}{dx} = 2x + \frac{1}{\tan^{-1} x (1+x^2)} - \frac{x}{1+x^2}$$

$$\frac{dy}{dx} = \frac{e^{x^2} \tan^{-1} x}{\sqrt{1+x^2}} \left[2x + \frac{1}{\tan^{-1} x (1+x^2)} - \frac{x}{1+x^2} \right]$$